

Nikhil Nair

MSc Systems and Control
Delft University of Technology (TU Delft)

GO in

+31 0644164182

✉: www.nikhilnicky972@gmail.com

ABOUT ME

Robotics Software Engineer with **3+ years of experience** developing **C++**, **Python**, and **MATLAB/Simulink** systems across ROS 2, simulation, perception, planning, control, and sensor integration. Background includes **project-based software development** for industrial robotics, **autonomous systems**, and **research-driven applications**, with hands-on work in Docker, Linux, Git, Gazebo, MoveIt, point-cloud processing, sensor fusion, and custom hardware interfaces. Strong technical foundation in systems and control, including modelling, Kalman filtering, optimal control, MPC, reinforcement learning, **algorithm development**, **MATLAB-based simulation**, and **model-based development**. Experienced in **translating complex technical requirements into working software demonstrators**, validating systems through simulation and field testing, writing **technical documentation**, and collaborating with **scientists, engineers, and clients** to deliver practical **R&D software solutions**.

EDUCATION

Delft University of Technology	2023 - 2025
MSc Systems and Control	Grade: Cum Laude, Courses : 8.43/10, Thesis: 9.0/10
National Institute of Technology Karnataka	2018-2022
B.Tech in Electrical and Electronics Engineering, minor in Computer Science	CGPA 8.42/10

SKILLS SUMMARY

- **Languages:** C++, Python, C, MATLAB Simulink
- **Frameworks:** ROS 1 & 2, TensorFlow, PyTorch, NumPy
- **Tools & Platforms:** Git, Docker, Linux
- **Systems and Control:** Dynamical System Modelling, Kalman Filtering, Nonlinear Control, Optimal Control, Model Predictive Control, Reinforcement Learning

MASTER THESIS

Optimal Control of Soft Robots , <i>Delft University of Technology</i> 	September 2024 – September 2025
Daily Supervisor: Dr. Daniel Feliu Tategón	
Supervisors: Prof. Cosimo Della Santina, Prof. Tamás Keviczky	
<i>Investigating Differential Dynamic Programming (DDP) for continuum soft robots modeled with the Geometric Variable Strain (GVS) framework. Benchmarked DDP vs. Partial Feedback Linearization on a soft inverted pendulum and used analytical gradients (RNEA) to improve convergence.</i>	

WORK EXPERIENCE

Mechatronics Consultant	Nov 2025 - Present
<i>ALTEN Nederlad B.V, Netherlands (Client: Grimbergen Industrial Systems) (Full-time)</i>	
• Developing an end-to-end robotic welding workflow (perception → planning → execution) using ROS2, MoveIt2, and custom control interfaces within a Docker-based development environment (C++ and Python). • Ported legacy ROS1 industrial robot drivers to ROS2 and developed a custom ROS2 Control hardware interface (C++) for a FANUC 30iB controller using TCP/IP Ethernet communication. • Integrating Zivid 3D camera and industrial laser scanner for perception-driven welding applications, implementing point cloud processing pipelines using Open3D and PCL (plane detection, edge detection, weld seam localization). • Conducting feasibility analysis and literature review on hydraulic excavator automation, including sensor retrofitting strategies, control architectures, and motion planning for closed-loop kinematic chains. • Designing and implementing a full ROS2-based simulation pipeline for excavator automation using Gazebo and MoveIt2, including modeling of closed-chain linkages and development of custom inverse kinematics solutions.	
Visiting Robotics Engineer	May 2025 - Aug 2025
<i>Eternal.ag, Germany (Part-time)</i>	
• Contributed to the deployment and field testing of a greenhouse mobile manipulation platform combining a custom mobile base with a UR5e manipulator. • Supported integration and debugging of the MoveIt-based manipulation stack and perception pipeline, resolving calibration, planning, and execution issues during on-site deployment. • Developed a ROS2 driver in C++ for a custom mobile base using CANopen motor controllers, implementing PDO/SDO communication and object dictionary mapping to enable low-level actuation and real-time odometry integration. • Worked within a Docker-based development environment using C++ and Python, integrating GPU-accelerated components from the NVIDIA Isaac robotics stack (including cuMotion and nvBlox) for perception and motion planning.	

Acceleration Robotics, Pune Client: DRDO-CAIR, Bengaluru, India

- Led development of multi-domain simulation environments (ground, aerial, and underwater vehicles) in Gazebo (Ignition/Gazebo Garden) to validate autonomy stacks before hardware testing.
- Ported simulation features and plugins from Gazebo Classic to modern Gazebo (Ignition), developing custom C++ model and sensor plugins to replicate realistic sensor behavior.
- Architected and developed a plugin-based ROS 2 sensor fusion framework in C++ built on the ANYbotics GridMap library, enabling multi-layer environmental representation with circular buffer storage.
- Designed and implemented fusion algorithms combining 3D LiDAR and camera data into unified costmaps for navigation and perception applications.
- Served as primary technical interface for the client, translating system requirements into deployable ROS 2 and simulation solutions.

PUBLICATIONS

- **N. Nair**, D. Caradonna, A. T. Mathew, F. Renda, C. Della Santina and D. Feliu-Talegón, "Stabilizing Low-Stiffness Soft Robots at Unstable Equilibria: A Model-Based Optimal Control Approach," *to appear in Proc. 2026 IEEE 65th Conference on Decision and Control (CDC)*, Dec. 2026. [Accepted]
- D. Caradonna, **N. Nair**, A. T. Mathew, D. Feliu-Talegón, I. Afgan, E. Falotico, C. Della Santina and F. Renda, "Soft Swing-up: Benchmarking Model-Based Optimal Control for Rigid-Soft Underactuated Systems," *arXiv preprint*, 2026. [Submitted to IEEE Robotics and Automation Letters (RAL)]
- R. Muduli, **N. Nair**, S. Kulkarni, M. Singhal, D. Jena and T. Moger, "Load Frequency Control of Two-area Power System Using an Actor-Critic Reinforcement Learning Method-based Adaptive PID Controller," *2023 IEEE 3rd International Conference on Sustainable Energy and Future Electric Transportation (SEFET)*, Bhubaneswar, India, 2023, pp. 1-6.

CONTROLS & PLANNING COURSEWORK

- **Model Predictive Control (SC42125)**: Implemented constrained MPC for an underactuated wheeled acrobat, including linearization, discretization, reference tracking, and output-feedback MPC.
- **Integration Project (SC42035)**: Modeled and identified a cart-pole system, implemented Luenberger/Kalman observers, and compared LQR vs. MPC control performance.
- **Robust Control (SC42145)**: Designed SISO/MIMO controllers for a floating wind turbine using mixed-sensitivity H_∞ design; evaluated tracking and disturbance rejection performance.
- **Hybrid Systems (SC42075)**: Built a hybrid automaton for a compass-gait walker and developed MPC for adaptive cruise control using MLD modeling and explicit MPC.
- **Control Engineering (SC42095)**: Designed continuous and discrete controllers for satellite attitude control using pole placement, LQ control, and observer-based output feedback under actuator limits.

SELECTED PROJECTS

Janitorial Cleaning Robot

Jan 2022 – Apr 2022

ARTPARK Robotics Challenge, IISc Bangalore

Designed and built a ROS1-based janitorial robot capable of autonomous washroom cleaning tasks including trash detection, washbasin sanitization, and floor mopping.

- Built autonomous navigation with LiDAR and depth camera, and integrated a YOLOv4-based object detection pipeline.
- Built a holonomic driver base with a custom made 6DOF arm. Did PCB design for the electronic subsystem for the base to interface with the motor drivers, Teensy, BMS.
- Arm was built with DYNAMIXEL servos daisy chained;
- Delivered a Gazebo simulation ahead of the hardware round and contributed to system bring-up under a \$5000 budget.

Automatic Generation Control Using an Adaptive PID controller

January 2022 - April 2022

Bachelor Thesis, National Institute of Technology, Surathkal, Karnataka

This project presents an Actor-Critic Reinforcement Learning-based adaptive PID controller for Automatic Generation Control (AGC). Unlike model-based methods that depend on accurate plant models, this approach updates PID parameters in real time, making it more adaptable to varying system configurations.

- Implemented and evaluated control performance under varying system conditions.